PL model:

Consider independent data noise(depolarizing) and syndrome noise (bit flips).
After measurement we have:

$$z = H_0 e + e_z$$

where: $z \in \mathbb{F}_2^{m\times 1}$ is observed syndrome
$e_z \in \mathbb{F}_2^{m\times 1}$ is syndrom noise
$H_0 \in \mathbb{F}_4^{m\times n}$ is check matrix, not necessarily full-rank
$e \in \mathbb{F}_4^{n\times 1}$ is data errors, corresponding to Pauli errors
Note that the product of H_0 and e elements is carried out over trace inner product

To decode e and e_z jointly. which is shown to perform better, we can build extended check matrix
write:

$$z = (H_0 \;\; I_m)\begin{pmatrix} e \\ e_z \end{pmatrix}$$

← original problem

However, $(H_0 \; I_m)$ gives most likely also a bad Tanner graph

We can do further transformation, write

$$z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} H & I_{m-r} & 0 \\ RH & 0 & I_r \end{pmatrix}\begin{pmatrix} e \\ e_{z1} \\ e_{z2} \end{pmatrix}$$

where $r = \mathrm{rank}_{\mathbb{F}_2}(H_0)$ is the number of stabilizer generators, and
$z_1 \in \mathbb{F}_2^{(m-r)\times 1}$, $z_2 \in \mathbb{F}_2^{r\times 1}$.

$\begin{pmatrix} H \\ RH \end{pmatrix} = H_0$ with $\mathrm{rank}_{\mathbb{F}_2} H = r$ and $H \in \mathbb{F}_4^{r\times n}$

$RH \in \mathbb{F}_4^{(m-r)\times n}$ is the redundant linearly dependent part
over $\mathbb{F}_2$

$R \in \mathbb{F}_2^{(m-r)\times r}$ is the linear transform matrix.

$e_{z1} \in \mathbb{F}_2^{(m-r)\times 1}$, $e_{z2} \in \mathbb{F}_2^{r\times 1}$
syndrome noise for full-rank part   syndrome noise for redundant part

Now we find ways to improve the graph connectivity: ← decoder optimization

Approach 1:

From:

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Quantum Data-Syndrome Codes

Alexei Ashikhmin, *Fellow, IEEE*, Ching-Yi Lai, *Member, IEEE*, and Todd A. Brun, *Senior Member, IEEE*

$$\cdot R \curvearrowright \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} H & I_{m-r} & 0 \\ RH & 0 & I_r \end{pmatrix}\begin{pmatrix} e \\ e_{z1} \\ e_{z2} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} z_1 \\ Rz_1 + z_2 \end{pmatrix} = \begin{pmatrix} H & I_{m-r} & 0 \\ 0 & R & I_r \end{pmatrix}\begin{pmatrix} e \\ e_{z1} \\ e_{z2} \end{pmatrix}$$

Approach 2:

From Classical product code constructions for quantum Calderbank-Shor-Steane codes

Dimiter Ostrev, Davide Orsucci, Francisco Lázaro, and Balazs Matuz

First find matrix M such that $MH_0 = 0$.
Then, we have

$$MH_0 e = 0 \qquad ①$$

$$\Rightarrow \begin{pmatrix} MH_0 & 0 \end{pmatrix} \begin{pmatrix} e \\ e_z \end{pmatrix} = 0 \qquad ②$$

we can append ② to ③

$$\begin{pmatrix} H_0 & I_m \end{pmatrix} \begin{pmatrix} e \\ e_z \end{pmatrix} = z \qquad ③$$

Then, we have

$$\begin{pmatrix} H_0 & I_m \\ MH_0 & 0 \end{pmatrix} \begin{pmatrix} e \\ e_z \end{pmatrix} = \begin{pmatrix} z \\ 0 \end{pmatrix}$$

Multiply the first row with M and add it to second row

$$\underbrace{\begin{pmatrix} H_0 & I_m \\ 0 & M \end{pmatrix}}_{:=\tilde{H},\ \text{known}} \underbrace{\begin{pmatrix} e \\ e_z \end{pmatrix}}_{:=\tilde{e}} = \underbrace{\begin{pmatrix} z \\ z+Mz \end{pmatrix}}_{:=\tilde{z},\ \text{known}}$$

We can BP decoding as usual for $\tilde{H}\tilde{e} = \tilde{z}$ with known $\tilde{H}$ and $\tilde{z}$. find $\tilde{e}$.

Compare approach 1 and 2:

In 1:

$$\begin{pmatrix} z_1 \\ Rz_1 + z_2 \end{pmatrix} = \begin{pmatrix} H & I_{m-r} & 0 \\ 0 & R & I_r \end{pmatrix} \begin{pmatrix} e \\ e_{z_1} \\ e_{z_2} \end{pmatrix} \qquad (a)$$

For 2, we can rewrite a bit:

$$\begin{pmatrix} H_0 & I_m \\ 0 & M \end{pmatrix} \begin{pmatrix} e \\ e_z \end{pmatrix} = \begin{pmatrix} z \\ z+Mz \end{pmatrix}$$

$$\Rightarrow \left(\begin{array}{c|cc} H & I_{m-r} & 0 \\ RH & 0 & I_r \\ \hline 0 & \multicolumn{2}{c}{M} \end{array}\right) \begin{pmatrix} e \\ e_{z_1} \\ e_{z_2} \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \\ z+Mz \end{pmatrix} \qquad (b)$$

Comparing (a) and (b) we see that the information at hand is the same.

The difference is that (b) contains more redundant equations (constraints), which may improve BP performance, depending on the structure of M.

Above derivation is based on the so-called "single-shot" decoding
But we could also consider multiple rounds of measurement with/without redundant syndrome measurements, see, e.g.,

Generalized quantum data-syndrome codes and belief propagation decoding for phenomenological noise

Kao-Yueh Kuo and Ching-Yi Lai